15/10/26

Signal Processing

- Recap: LTI system analysis using LCCDE - Causality, stability, Inverse systems
- Inverse Systems:
  - $H(z) = (1 - 0.5z^{-1})/(1 - 0.9z^{-1})$, $|z| > 0.9$
  - $H(z) = (z^{-1} - 0.5)/(1 - 0.9z^{-1})$, $|z| > 0.9$
- All-pass Systems
- Minimum-phase Systems
- Inverse Systems Revisited
  - $H(z) = (1 + 3z^{-1})/(1 + \frac{1}{2}z^{-1})$
  - $H(z) = \left[\left(1 + \frac{2}{2}e^{j\pi/4}z^{-1}\right)\left(1 + \frac{3}{2}e^{-j\pi/4}z^{-1}\right)\right]/\left(1 - \frac{1}{3}z^{-1}\right)$
- Recap: $y[n] - \frac{5}{2}y[n-1] + y[n-2] = x[n]$ ⟹

$$H(z) = k/\left[\left(1 - \tfrac{1}{2}z^{-1}\right)\left(1 - 2z^{-1}\right)\right]$$; 3 possible regions of convergence

- Inverse systems: $H(z) = (1 - 0.5z^{-1})/(1 - 0.9z^{-1})$; $|z| > 0.9$

  $H_{inv}(z) = (1 - 0.9z^{-1})/(1 - 0.5z^{-1})$; $|z| > 0.5$

  $H_{inv}'(z) =$ ——"—— ; $|z| < 0.5$?
- $H(z) = \left(\dfrac{z^{-1} - 0.5}{1 - 0.9z^{-1}}\right)$; $|z| > 0.9$ $\quad H(z) = \dfrac{(2z^{-1} - 1)}{2(1 - 0.9z^{-1})}$, $|z| > 0.9$
- $H_{inv}(z) = 2 \cdot \dfrac{(1 - 0.9z^{-1})}{(2z^{-1} - 1)}$; $|z| > 2$.

  $R_H \cap R_{H_{inv}}$ is $|z| > 2$. H_{inv} is causal, unstable,
- $|z| < 2$; $R_H \cap R_{H_{inv}}$ $\quad 0.9 < |z| < 2$; H_{inv} is non causal, stable

- We are interested in causal and stable systems

- If $H(z) = \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=0}^{N} (1 - c_k z^{-1})}{\prod_{k=0}^{M} (1 - d_k z^{-1})}$, a causal inverse of this system has

$$\text{ROC} \quad |z| > \max_k |c_k|$$

- Additionally, stability requires

$$\max_k |c_k| < 1$$

- Min-phase systems: An LTI system whose poles and zeros lie inside the unit circle.

- All-pass systems: $H_{ap}(z) = \frac{z^{-1} - a^*}{1 - a z^{-1}}$

$$H_{ap}(e^{j\omega}) = H_{ap}(z)\Big|_{z = e^{j\omega}}$$

$$= \frac{e^{-j\omega} - a^*}{1 - a e^{-j\omega}} = e^{-j\omega} \frac{(1 - a^* e^{j\omega})}{(1 - a e^{-j\omega})}$$

$$\Rightarrow |H_{ap}(e^{j\omega})| = \left| e^{-j\omega} \cdot \frac{(1 - a^* e^{j\omega})}{(1 - a e^{-j\omega})} \right|$$

$$= |e^{-j\omega}| \left| \frac{1 - w^*}{1 - w} \right| \quad \text{where } w = a e^{-j\omega}$$

$$= 1 \cdot 1.$$

Q. How do all-pass systems help?

A. In finding the inverse of causal and stable systems

- Inverting causal & stable systems
  - Recall: Causality requires the ROC to be outside the largest pole.
    - Stability requires the ROC to include $|z| = 1 \Rightarrow$ largest pole mag < 1
    - However, causality and stability impose no restrictions on the location of zeros
- Let's look at an arbitrary causal and stable system $H(z)$ with a zero at $1/c^*$ outside the unit circle.

Let $H(z) = H_{min}(z) \cdot (z^{-1} - c^*)$

$$= H_{min}(z) \cdot (z^{-1} - c^*) \frac{(1 - cz^{-1})}{(1 - cz^{-1})}$$

$$= \underbrace{H_{min}(z) \cdot (1 - cz^{-1})}_{\text{min phase}} \cdot \underbrace{\frac{(z^{-1} - c^*)}{(1 - cz^{-1})}}_{\text{All-pass}}$$

$$H(z) = H'_{min}(z) \cdot H_{ap}(z)$$

Let $H'_{inv}(z) = \frac{1}{H'_{min}(z)}$

$$\Rightarrow \left| H(z) \cdot H'_{inv}(z) \right|_{z = e^{j\omega}} = \left| H'_{min}(z) \cdot H_{ap}(z) \cdot \frac{1}{H'_{min}(z)} \right|$$

$$= \left| H_{ap}(z) \right|_{z = e^{j\omega}}$$

$$= 1$$